COMPUTER ENGINEERING

QUEZON CITY

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| **COURSE CODE/TITLE:** | CPE010 | **SECTION:** | CPE21S4 |
| **DATE PERFORMED:** | 01/10/2026 | **DATE SUBMITTED:** | 01/10/2026 |
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**ACTIVITY NO. 10.1** **Graph**

**1. PROCEDURE OUTPUT**

What procedures did you perform?

## General Instructions:

1. Define the following: (10)

A. Graph
A graph is a group of nodes connected by edges.

B. Vertex
A vertex is a point in a graph. It is also called a node.

C. Nodes
Nodes are the points or objects in a graph.

D. Weight
Weight is the number or value given to an edge.

E. Path
A path is the way from one node to another node.

F. Directed Graph
A directed graph is a graph where the edges have arrows showing the direction.

G. Connected Graph
A connected graph is a graph where the nodes are connected to each other.

H. Directed Cyclic
It is a directed graph that has a path that can go back to the starting node.

I. Adjacency Matrix
An adjacency matrix is a table that shows which nodes are connected.

J. Adjacency List
An adjacency list shows each node and the other nodes connected to it.

2. Identify the parts of the following graph. (10)



Nodes:

A B C D

Edges:

A → B = 3, A → C = 2, C → B = 1, D → A = 4

Number of nodes: 4

Number of edges: 4

The graph is directed because the edges have arrows.

3. Plot the adjacency matrix of the graph above. (10 pts)

A B C D

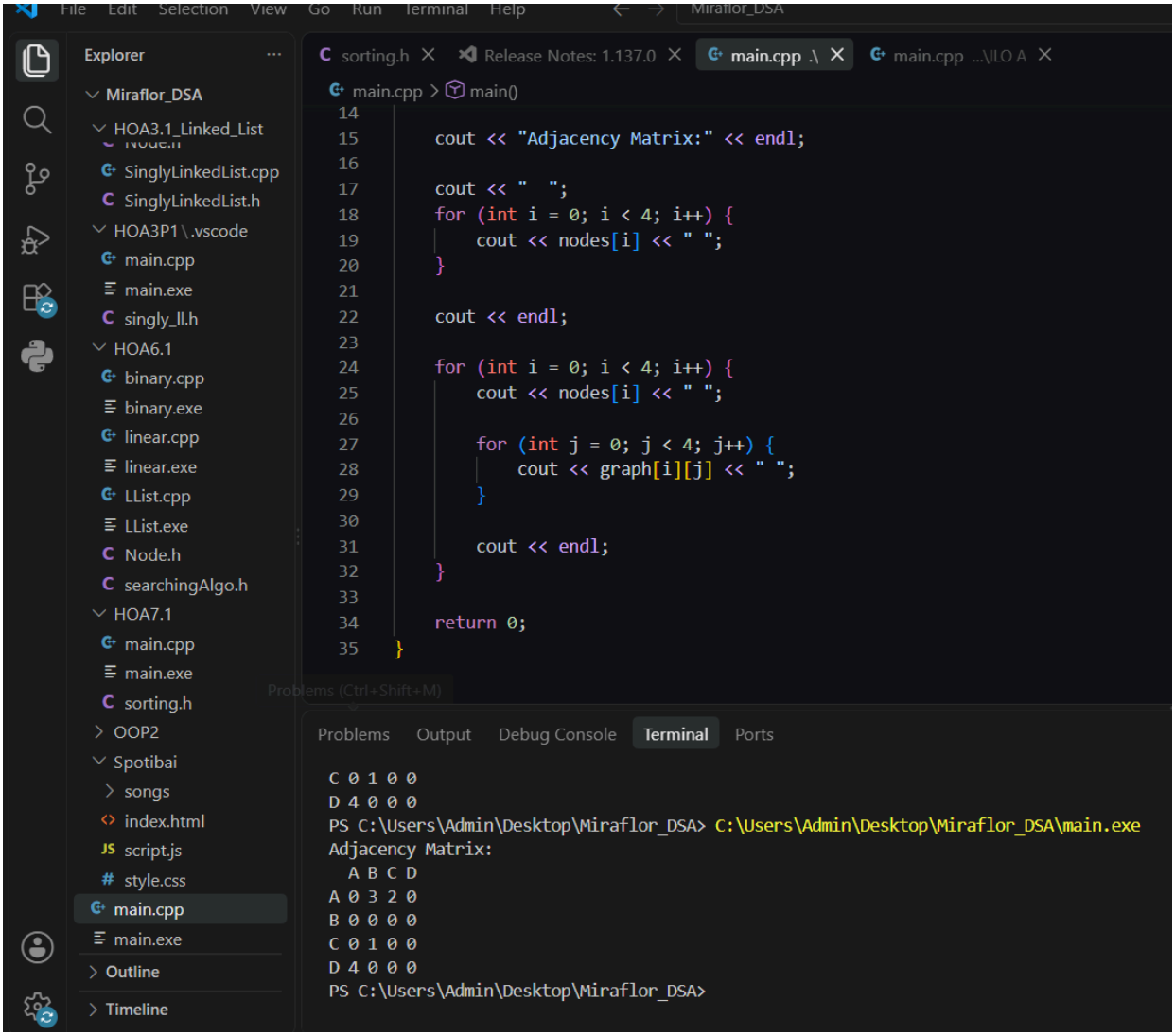
0 3 2 0

0 0 0 0

0 1 0 0

4 0 0 0

4. Create a program in C++ program to create a  graph that looks like in the above figure.




5. Differentiate a BFS vs DFS types of search.

BFS (Breadth First Search) BFS searches the graph level by level. It uses a queue.

DFS (Depth First Search) DFS goes deep into one path first before going back. It uses a stack or recursion.

Difference: BFS checks the nearby nodes first, while DFS follows one path as far as it can before going back.

## 2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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*Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.*

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